

1. The Data Synthesis & Python-PPTX Code Test

"Write a complete, error-free Python script using the `python-pptx` library to generate and save an 8-slide Quarterly Earnings deck named 'Q4_Earnings.pptx' for a SaaS company. The raw data is: Q1 revenue \$2.4M, Q2 \$2.8M, Q3 \$3.1M, Q4 \$2.9M. Customer churn increased by 4% in Q4. The script must programmatically generate a native PowerPoint bar chart on Slide 3 comparing the quarterly revenue, and a pie chart on Slide 5 showing a breakdown of churn. Include a 'Key Takeaway' text box on every slide. Ensure the script includes robust error handling and writes exact CFO speaker notes addressing the Q4 revenue dip."

2. The Direct File Output & Progressive Disclosure Test

"Use your file-generation tools to directly create and output a downloadable 10-slide `.pptx` file explaining Quantum Computing to high school students. You must use the visual concept of a 'maze' as a continuous metaphor. Start with basic concepts on slide 1 and progressively introduce complex concepts like superposition and entanglement by slide 6. Do not use any bullet points. Instead, structure the slide layouts dynamically (e.g., using two text boxes side-by-side, or top-heavy title layouts). Include a hyperlinked interactive 'Quiz' button on slides 4 and 8 that visually stands out."

3. The Strict RGB & Corporate Brand Code Test

"Write a Python script using `python-pptx` that generates a 6-slide product launch pitch deck saved as 'Launch.pptx'. The code must strictly adhere to these brand guidelines: The title text on every slide must be set to the exact RGB equivalent of Hex #1A365D, and all body text to Hex #E2E8F0. The font must be set to Arial. You are strictly forbidden from using the words 'synergy', 'disruptive', 'revolutionary', or 'innovative'. Every slide must have a footer that says 'Confidential - Internal Use Only'. Slide 3 must contain a programmatically generated 3x4 PowerPoint table comparing three pricing tiers."

4. The Multilingual Layout & Localization File Test

"Generate and output a downloadable 5-slide market entry strategy `.pptx` file for expanding a US-based retail brand into the United Arab Emirates. The presentation must be strictly bilingual. On every single slide, you must position the English text in a placeholder on the left half of the slide, and the culturally localized Arabic translation in a placeholder on the right half. Ensure the Arabic text is right-to-left (RTL) formatted if your generation tool supports it. Include speaker notes in English only, explaining the cultural rationale behind the Arabic phrasing chosen."

5. The VBA Macro Execution Test

"I do not want a Python script or a generated file. I want you to write the complete, executable VBA (Visual Basic for Applications) macro code that, when run inside PowerPoint, will automatically generate a 4-slide presentation about the 'History of Artificial Intelligence'. Slide 1 must be a title slide. Slide 2 must contain a timeline drawn using basic PowerPoint shapes (Rectangles and Connectors) positioned via exact VBA coordinates. Slide 4 must contain a perfectly centered text box that says 'The Future'. The VBA must set the background color of all slides to dark gray (RGB 64, 64, 64) and text to white."

6. The Narrative Arc Code Generation Test

"Write the complete Python code to generate a 12-slide investor pitch deck (`pitch.pptx`) for a sustainable agriculture startup. Structure the content exactly according to the 'Hero's Journey' narrative framework. The 'Hero' is the modern farmer, the 'Villain' is supply chain inefficiency, and our startup is the 'Magic Weapon'. Your code must explicitly label which stage of the Hero's Journey each slide represents in the slide title. Furthermore, the script must apply specific slide transitions (e.g., Fade, Push) to each slide if the library supports it, matching the emotional tone of that story stage."

7. The Interactive Branching Navigation Test

"Use your file-generation tools to output a fully functioning 7-slide `.pptx` HR compliance training presentation on workplace safety. This deck must act as an interactive quiz. Slide 2 presents a safety hazard scenario with three text boxes acting as buttons. You must programmatically set up internal hyperlinks for these text boxes: Option A must link to Slide 3 (Incorrect feedback), Option B must link to Slide 4 (Incorrect feedback), and Option C must link to Slide 5 (Correct feedback). Include a 'Back to Question' hyperlinked button on slides 3, 4, and 5."

8. The Academic Formatting & Table Generation Test

"Write a Python script using `python-pptx` to create a 9-slide academic literature review (`CRISPR_Review.pptx`) on the efficacy of CRISPR-Cas9 in agriculture. You must cite at least four distinct real-world scientific papers. Every slide must include proper APA formatted in-text citations. The script must generate a complex 5x4 matrix table on Slide 7 comparing the methodologies of the four papers, ensuring the column widths are evenly distributed. Slide 9 must be a perfectly formatted APA reference list. Populate the speaker notes with extensive scientific context."

9. The Crisis Management & Constraints File Test

"Generate and output an 8-slide PR Crisis Management `.pptx` file for an airline that just experienced a major data breach. The file must strictly follow these constraints: Every slide must include a text box at the absolute bottom with a

legally approved disclaimer reading 'Subject to Ongoing Investigation - Privileged and Confidential'. The speaker notes must be divided into two formatted sections: 'What we know (Internal Fact)' and 'What we are legally cleared to say (External Statement)'. Do not include any images, icons, or decorative shapes; the layout must be purely text-based and somber."

10. The Multi-Agent Persona Code Test

"Write a Python script that builds a 6-slide .pptx presentation proposing a highly experimental AI marketing campaign. You must act as two conflicting experts: a visionary CMO and a conservative CRO. The code must generate the slide text representing their final compromise. However, the code must populate the speaker notes of each slide with a formatted dialogue transcript showing the CMO and CRO debating what should go on the slide. The script must end by generating a 2x2 'Risk vs. Reward' matrix using intersecting line shapes and text boxes on Slide 6."